# MALDI Biotyper ASTM Communication

*Interface Specification Version 4.5*

## Document History

| Date | Version | Author | Remarks |
|---|---|---|---|
| 2009-04-16 | 1.0 | O. Rombauts (MIPS) | Initial revision |
| 2009-09-24 | 2.0 | O. Rombauts (MIPS) | Update result export, add patient query |
| 2011-07-18 | 3.0 | K.-O. Kräuter (Bruker) | Extend result sending definition, correct patient query, add measuring task queries and further generalizations |
| 2011-09-30 | 3.0.1 | K.-O. Kräuter (Bruker) | Added missing frame numbers in result example (3.1.1.3) |
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| 2019-05-16 | 4.3 | K.-O. Kräuter (Bruker) | Added fields for the sample and run UUIDs (P.3.1.1 and P.3.1.2) in 3.1.1.2.<br>Added fields for specific subtyping information in the Result Message (R.4.1.8 and R.4.1.9) in 3.1.1.2. |
| 2021-09-01 | 4.4 | K.-O. Kräuter (Bruker) | Added metadata to header record (H.4, H.5, H.6) in 3.1.1.2. |
| 2023-07-10 | 4.5 | K.-O. Kräuter (Bruker) | Added optional mixed culture info (ORG Result: R.4.1.4) in 3.1.1.2. |

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# 1 Introduction

This document describes the ASTM interface used for connecting the MALDI Biotyper (MBT) to a laboratory information management system (LIMS). The first LIMS connected to the MBT via ASTM was GLIMS from the Belgian provider MIPS. Therefore the first version of this document has been created by MIPS and has been maintained in close corporation with this LIMS provider.

Currently several LIMSes talk to the MBT via ASTM as well as some AST (antibiotic susceptibility testing) automates like from BD, Siemens or Trek.

Although the ASTM interface definitions are more than 20 years old and have meanwhile been replaced by newer standards (e.g. HL7), such interfaces are still found in many laboratories and probably will be for the next years. Therefore it is still required also for a modern laboratory instrument like the MBT to *speak* the ASTM protocol.

If you look into the appropriate ASTM specification document named *"Standard Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems"*, you will recognize that this protocol was initially created for a serial line communication. But the serial line specifics are so well separated from the general protocol definition, that the complete transport layer may completely be exchanged. In the case of the MBT this has been done. The serial line connection has been replaced by a TCP/IP based connection.

A more detailed definition about what data is sent and what is received when the MBT and a LIMS are talking to each other is given in the next paragraph.

# 1 Supported Workflows

There are two main types of supported workflows:

1. The MBT is ***sending*** results to the LIMS.
2. The MBT is ***querying*** the LIMS for specific information.

Sending ***results*** from the MBT to the LIMS is the easiest and also most important workflow. The results sent to the LIMS consist of the found specimen name or code together with the identifier (accession number + isolation) for every sample measured in the MBT. Due to the way the MBT is working there is always a result but with a higher or lower reliability. Therefore a reliability number (score values) is also provided for each result. The ASTM message specification for sending MBT results is presented in paragraph ***3.1.1*** (***Result Message)***.

There are several ways for the MBT to query information from the LIMS. Historically first was the ***query for patient data***. No patient data is available in the MBT database for data security reasons. But there are situations (e.g. result validation) where the MBT user would like to access patient data for a certain sample. To solve this dilemma a patient data query is provided by the LIMS where the MBT uses an appropriate sample identifier (accession number + isolation). The received patient data is just displayed in the MBT graphical user interface and is never saved. The ASTM messages for patient data queries are presented in paragraph ***3.2*** (***Patient Query***).

The other data query introduced with version 3.0 of the ASTM communication document is the ***query for measuring tasks***. A measuring task is closely related to a prepared MALDI target plate. It specifies what samples have been prepared on which target plate positions (also called *wells*). So this list of prepared positions is identical to the plate positions to be measured. The sample identifiers from this list are finally connected to the results gained at these positions.

In the most simple workflow the creation of the measuring tasks is completely done inside the MBT application (MBT-RTC wizard). In this case no measurement task query is required. But if the measurement tasks are created outside the MBT (e.g. in the LIMS) the appropriate query may be used

to import a measurement task into the MBT. There are two alternative parameters used in such a query. Either just the name of the requested measuring task is provided or the identifier of the MALDI target plate for which the measuring task has been defined (bar-coded workflow).

The ASTM messages for measuring task queries are presented in paragraph ***3.3*** (***Measuring Task Query***). The ASTM messages used to query for a measuring task list are presented in paragraph ***3.4*** (***Measuring Task List Query***).

# 2 Communication

The central documents describing the communication are the ASTM standards ***E 1394–97*** (high level) and ***E 1381–95*** (low level). The first one defines how the data cargo load is distributed into ASTM specific compartment (fields and records in ASTM frames). While the second one specifies how the so prepared strings are decorated with control code and sent over the communication channel, which in our case is a TCP socket connection.

The *low level protocol specification* (***E 1381–95***) is strictly observed, because this is the basis for a reliable data transfer. The high level specification (***E 1394–97***) on the other hand is interpreted more laxly, not least because there are no MALDI-TOF related ASTM fields in the specification anyway. Therefore it is crucial to refer to this document about which data is represented where in the ASTM structure.

Finally it is important to notice that the communication channel between MBT and LIMS is not symmetric. The LIMS always acts as the listener waiting for the MBT to start the communication. For the used TCP sockets this corresponds to the LIMS being the socket server while the MBT is the socket client.

Accordingly opening the channel starts with the socket client (MBT) requesting a connection from the socket server (LIMS) listening on a certain host at a certain port (TCP). After the channel has been successfully opened the data transfer proceeds according to the ASTM low level protocol specification. The client starts by sending the enquiry character (usually symbolized by **<ENQ>**) and waits for the acknowledge character (symbolized by **<ACK>**) from the server. Then each ASTM frame is sent over in the defined format, checked by the server and answered with **<ACK>** if valid. After the last frame has been sent (or the transfer was ended for other reasons) the client sends the end-of-transmission character (symbolized by **<EOT>**) and immediately closes the channel (without waiting for a **<ACK>** from the server). This means that the channel is not kept open until the next data transfer is due. It is rather opened every time again when required.

A complete but concise communication example can be found in paragraph ***3.1.1.3***.

# 3 Message Structures

## 3.1 Result Message Overview

The ASTM result message is used for sending MBT results to the LIMS. The patient information field just contains a counter and no real patient data as this data resides in the LIMS only. The square brackets [..] in the structure definition below indicate repeatable blocks.

### 3.1.1 Result Message

#### 3.1.1.1 Message Structure

**Message header**
**[**
**Patient Information**
**[**
**Specimen Information**
**[**
**Isolation ORG – result**
**[**
**Isolation PRO – result**
**]**
**]**
**]**
**]**
**Terminator**

#### 3.1.1.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | `H\|\^&\|\|M1\|M2\|M3<CR>` | Standard ASTM start |
| H.2 | Delimiters | `\|\^&` | Standard delimiters |
| H.4 | User Name + Host Name | **`Creator field from run metadata`** | Optional metadata |
| H.5 | Run Name | **`Name field from run metadata`** | Optional metadata |
| H.6 | Run Description | **`Description field from run metadata`** | Optional metadata |
| | ***Patient information*** | `P\|` Seq. No `\|`Run UUID `^` Sample UUID `<CR>` | |
| P.2.1.1 | Seq. No | Number of P-record | Numbering the samples |
| P.3.1.1 | Run UUID | Universally Unique Identifier of the run | This UUID is identical for all samples measured in one run. |
| P.3.1.2 | Sample UUID | Universally Unique Identifier of the sample | This UUID is unique for each sample. |
| | ***Specimen information*** | `O\|` Seq. No `\|` Accession number `^` Accession | |

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | | name ^^ Measurement date-time ^^ Instrument Id ^ Target plate Id ^ Target plate position<CR> | |
| O.2.1.1 | Seq. No | Number of O-record | |
| O.3.1.1 | Accession number | Sample identifier (for Isolation see R.3.1.5) | Required Parameter |
| O.3.1.2 | Accession name | Sample name | Optional Parameter |
| O.3.1.4 | Measurement date-time | Time stamp of measurement using the format: YYYY-MM-DD[T\| ]hh:mm:ss.s | MBT server 3.0+ replaces *blank* between date and time with a T |
| O.3.1.6 | Instrument Id | MALDI-TOF instrument identifier | Optional parameter |
| O.3.1.7 | Target plate Id | MALDI target plate identifier | Optional parameter |
| O.3.1.8 | Target plate position | MALDI target plate position | Optional parameter (e.g. **A1**) |
| | ***Isolation ORG result*** | R\| Seq. No \|^^^ORG^ Isolation sequence number \| Organism code ^^ Consistency category ^^ Project type ^ Sample type ^ Validation flag ^ Subtyping typed ^ Subtyping presumptive <CR> | |
| R.2.1.1 | Seq. No | Number of R-record | |
| R.3.1.5 | Isolation sequence number | Isolation number (if required) | Defaults to 1 |
| R.4.1.1 | Organism code | Organism code of the identified isolation | Lab specific look-up |
| R.4.1.3 | Consistency | A\|B\|C (high \| low \| unknown) | Generated from the hit list of one result |
| R.4.1.4 | Mix culture info | <optional-prefix>Genus Species | Genus Species name of second mixture component |
| R.4.1.5 | Project type | RUO\|IVD (research-use-only \| in-vitro-diagnostic) | Optional parameter (if not present the type **RUO** is assumed) |
| R.4.1.6 | Sample type | STD\|QC\|BC\|MYC\|FUNGI (standard \| quality control \| sepsityper \| mycobacteria \| filamentous fungi) | Optional parameter (if not present the type **STD** is assumed) |
| R.4.1.7 | Validation flag | (empty)\|1 | Optional parameter (**1** means the result was validated in the MBT and the LIMS export was triggered manually; the default is *(empty)* – meaning automatically exported to the LIMS) |
| R.4.1.8 | Subtyping typed | Type information resulting from subtyping step (e.g. L. monocytogenes). | This type may supersedes the species from the MBT identification result. If no appropriate information is available this field stays empty. |
| R.4.1.9 | Subtyping presumptive | Presumptive type information from the subtyping step (e.g. L. innocua). | A presumptive type is considered as hint (“this could be the presumptive type”). If no appropriate |

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | | | information is available this field stays empty. |
| | ***Probability PRO result*** | R\| Seq. No \|^^^PRO^ Isolation sequence number ^ Isolation match number \| Organism code ^ Organism name ^ Qualitative probability ^ Quantitative probability ^ Matched pattern UUID ^ Matched pattern library<CR> | |
| R.2.1.1 | Seq. No | Number of R-record | Starting at 1 |
| R.3.1.5 | Isolation sequence number | Always 1 | not used |
| R.3.1.6 | Isolation match number | Counts the matches in the hit list | Starting at 1 |
| R.4.1.1 | Organism code | Organism code of the identified isolation | Lab specific look-up |
| R.4.1.2 | Organism name | Species name of matched specimen | <genus> <species> |
| R.4.1.3 | Qualitative probability | +++\|++\|+\|- | green\|green\|yellow\|red |
| R.4.1.4 | Quantitative probability | 0 ≤ Score value ≤ 3.0 | Negative score values are rounded up to 0 |
| R.4.1.5 | Matched pattern UUID | Universal unique Id of matching pattern from the reference database | Optional parameter |
| R.4.1.6 | Matched pattern library | MSP library name the appropriate reference pattern belongs to (if any). | Optional parameter (empty if not used) |
| | ***Terminator record*** | L \| 1 \| N <CR> | Standard ASTM end |

### 3.1.1.3 Communication Example (Result message)

In the following example several MBT results are sent to the LIMS.

Just the species of the best matching reference pattern (MSP) is sent (this can be configured in the LIMS export settings).

The ***Organism code*** used for this example is the NCBI taxonomy ID (e.g. ***562*** stands for ***E. coli***).

See: http://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Info&id=562

In the listing below data sent by the MBT are in blue while data sent by the LIMS are in red.


```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2P|1|d2668640-ca35-4074-b237-c5a2741d6d65^f29bf739-0aaf-4510-89d7-34d14631528b<CR><ETX>B3<CR><LF>
<ACK>
<STX>3O|1|1^B1^^2019-05-16T11:10:54.279^^269944.00824^101808002^B1<CR><ETX>4C<CR><LF>
<ACK>
<STX>4R|1|^^^ORG^1|1639^^A^^RUO^STD^1^L. monocytogenes^<CR><ETX>06<CR><LF>
<ACK>
<STX>5R|2|^^^PRO^1^1|1639^Listeria monocytogenes^++^2.23^7d4a866f-da3a-487d-99e8-30c696d614d3^BDAL<CR><ETX>82<CR><LF>
<ACK>
<STX>6P|2|d2668640-ca35-4074-b237-c5a2741d6d65^27a7ab51-4bc9-4368-b24e-3b8a1e228bd8<CR><ETX>3F<CR><LF>
<ACK>
<STX>7O|1|5^C6^^2019-05-16T11:10:54.295^^269944.00824^101808002^C6<CR><ETX>5E<CR><LF>
<ACK>
<STX>0R|1|^^^ORG^1|1639^^A^^RUO^STD^1^L. monocytogenes^<CR><ETX>02<CR><LF>
<ACK>
<STX>1R|2|^^^PRO^1^1|1639^Listeria monocytogenes^++^2.29^93f0b820-0591-454f-a0b0-9582334360d0^BDAL<CR><ETX>6C<CR><LF>
<ACK>
<STX>2P|3|d2668640-ca35-4074-b237-c5a2741d6d65^aea42202-f377-4612-ad93-760a60c8e288<CR><ETX>AC<CR><LF>
<ACK>
<STX>3O|1|6^C5^^2019-05-16T11:10:54.279^^269944.00824^101808002^C5<CR><ETX>5B<CR><LF>
<ACK>
<STX>4R|1|^^^ORG^1|1639^^B^^RUO^STD^1^L. monocytogenes^<CR><ETX>07<CR><LF>
<ACK>
<STX>5R|2|^^^PRO^1^1|1639^Listeria monocytogenes^++^2.16^93f0b820-0591-454f-a0b0-9582334360d0^BDAL<CR><ETX>6C<CR><LF>
<ACK>
<STX>6P|4|d2668640-ca35-4074-b237-c5a2741d6d65^a137b6cb-58cf-40c2-93eb-97df27d2865e<CR><ETX>81<CR><LF>
<ACK>
<STX>7O|1|7^C7^^2019-05-16T11:10:54.279^^269944.00824^101808002^C7<CR><ETX>64<CR><LF>
<ACK>
<STX>0R|1|^^^ORG^1|1639^^A^^RUO^STD^1^L. monocytogenes^<CR><ETX>02<CR><LF>
<ACK>
<STX>1R|2|^^^PRO^1^1|1639^Listeria monocytogenes^++^2.15^96a55a0a-29a8-4851-b029-858be553c563^BDAL<CR><ETX>DE<CR><LF>
<ACK>
<STX>2P|5|d2668640-ca35-4074-b237-c5a2741d6d65^aae350ca-a767-4da5-ac02-8dcf7e0278b0<CR><ETX>C2<CR><LF>
<ACK>
<STX>3O|1|8^C8^^2019-05-16T11:10:54.279^^269944.00824^101808002^C8<CR><ETX>63<CR><LF>
<ACK>
<STX>4R|1|^^^ORG^1|1639^^A^^RUO^STD^1^^<CR><ETX>E2<CR><LF>
<ACK>
<STX>5R|2|^^^PRO^1^1|1639^Listeria monocytogenes^++^2.25^93f0b820-0591-454f-a0b0-9582334360d0^BDAL<CR><ETX>6C<CR><LF>
<ACK>
<STX>6P|6|d2668640-ca35-4074-b237-c5a2741d6d65^b8c15fcf-f93e-4724-a3da-4311c071fb6d<CR><ETX>A0<CR><LF>
<ACK>
<STX>7O|1|9^C9^^2019-05-16T11:10:54.295^^269944.00824^101808002^C9<CR><ETX>68<CR><LF>
<ACK>
<STX>0R|1|^^^ORG^1|562^^A^^RUO^STD^1^^<CR><ETX>A8<CR><LF>
<ACK>
<STX>1R|2|^^^PRO^1^1|562^Escherichia coli^+++^2.68^631e1d31-81bd-4f97-b16b-195a5e43cfa9^BDAL<CR><ETX>CA<CR><LF>
<ACK>
<STX>2P|7|d2668640-ca35-4074-b237-c5a2741d6d65^edcbe46b-f5fc-4248-9192-2aaa6461868c<CR><ETX>7D<CR><LF>
<ACK>
<STX>3O|1|BTS^H12^^2019-05-16T11:10:54.295^^269944.00824^101808002^H12<CR><ETX>72<CR><LF>
<ACK>
<STX>4R|1|^^^ORG^1|562^^A^^RUO^QC^1^^<CR><ETX>55<CR><LF>
<ACK>
<STX>5R|2|^^^PRO^1^1|562^Escherichia coli^+++^2.45^5da4606c-708e-4678-985d-805bda7ce7b9^BDAL<CR><ETX>B2<CR><LF>
<ACK>
<STX>6L|1|N<CR><ETX>09<CR><LF>
<ACK>
<EOT>
```

## 3.2 Patient Query

### 3.2.1 Query Message

#### 3.2.1.1 Message Structure

**Message header**

**Request Information (Accession + Isolation)**

**Terminator**

#### 3.2.1.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | **H\|\\^&<CR>** | Standard ASTM start |
| H.2 | Delimiters | **\|\\^&** | Standard delimiters |
| | ***Request Information*** | **Q\|** Seq. No **\|^**Accession number **^** Isolation sequence number **\|\|\|\|\|\|\|\|\|\|** Request information status **<CR>** | |
| Q.2.1.1 | Seq. No | Number of the Q-record | Here always 1 |
| Q.3.1.2 | Accession number | | Specimen identifier |
| Q.3.1.3 | Isolation sequence number | | Isolation identifier |
| Q.13.1.1 | Request information status | **O** | not used |
| | ***Terminator record*** | **L \| 1 \| N <CR>** | Standard ASTM end |

#### 3.2.1.3 Communication Example (message query)

This is an actual query starting with the enquiry (**<ENQ>**) send by the MBT. Every line is acknowledged (**<ACK>**) by the LIMS. The query is finished by the MBT sending the end-of-transmission (**<EOT>**). Different from the result sending via ASTM as described in the previous chapter, the channel is not closed after the **<EOT>**. But the MBT waits for an enquiry now to be send from the LIMS (**<ENQ>**) thus starting to answer the query. The communication example containing the complete answer is listed in paragraph ***3.2.2.3***.

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2Q|1|^XY00012^1||||||||||O<CR><ETX>74<CR><LF>
<ACK>
<STX>3L|1|N<CR><ETX>06<CR><LF>
<ACK>
<EOT>
```

### 3.2.2 Query Result Message

#### 3.2.2.1 Message Structure

***When patient data is available for the queried specimen and isolate***

**Message header**

**Patient Information**

**Specimen Information (report type 'Q')**

**[**

**Isolation test ISOT – result request**

**]**

**Terminator**

***When no data is available for the queried specimen and isolate***

**Message Header**

**Patient Information**

**Specimen Information (report type 'Z')**

**Terminator**

***When queried specimen is not available***

**Message Header**

**Terminator**

### 3.2.2.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | H\|\^&<CR> | Standard ASTM start |
| H.2 | Delimiters | \|\^& | Standard delimiters |
| | ***Patient Information*** | P\| Seq. No \| Patient Id \| Second patient Id \|\| Last name ^ First name ^ Middle name ^^ Title \| Spouse name \| Birth \| Sex \|\|\|\|\|\|\|\| Physician code ^ Physician name \|\|\|\|\|\|\|\|\| Ward ^ Room \|\|\|\|\|\|\| Hospital service <CR> | |
| P.2.1.1 | Seq. No | Number of P-record | Starting at 1 |
| P.3.1.1 | Patient Id | | |
| P.4.1.1 | Second patient Id | | |
| P.6.1.1 | Last name | | |
| P.6.1.2 | First name | | |
| P.6.1.3 | Middle name | | |
| P.6.1.5 | Title | | |
| P.7.1.1 | Spouse name | | |
| P.8.1.1 | Birth | Format: **CCYYMMDD** | |
| P.9.1.1 | Sex | M\|F\|U | |
| P.17.1.1 | Physician code | | |
| P.17.1.2 | Physician name | | |
| P.26.1.1 | Ward | | |
| P.26.1.2 | Room | | |
| P.33.1.1 | Hospital Service | | |

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Specimen information*** | O\| Seq. No \| Accession number \|\|^^^ Test Id \|\| Requested time \| Collection time \|\|\|\| Action code \|\|\|\| Specimen type ^ Specimen source \|\|\|\|\|\|\|\|\|\| Report type <CR> | |
| O.2.1.1 | Seq. No | Number of O-record | Starting at 1 |
| O.3.1.1 | Accession number | | |
| O.5.1.4 | Test Id | Name of examination | |
| O.7.1.1 | Requested time | Time of order request | |
| O.8.1.1 | Collection time | Time of specimen collection | |
| O.12.1.1 | Action code | N | |
| O.16.1.1 | Specimen type | Specimens culture type like *blood*, *urine*, *stool*, *serum*, *wound*, *biopsy*, *sputum* etc. | |
| O.16.1.2 | Specimen source | Used to specify the specimen source (if required) like *left arm*, *left hand*, *right lung*. | |
| O.26.1.1 | Report type | Q\|Z | not used |
| | ***Isolation ORG result request*** | R\| Seq. No \|^^^ORG^ Isolation sequence number \| Organism code <CR> | |
| R.2.1.1 | Seq. No | Number of R-record | Starting at 1 |
| R.3.1.5 | Isolation sequence number | | |
| R.4.1.1 | Organism code | Organism code of isolation (if present) | |
| | ***Isolation test ISOT – result request*** | R\| Seq. No \|^^^ISOT^ Isolation sequence number \|^ Isolation test code <CR> | |
| R.2.1.1 | Seq. No | | |
| R.3.1.5 | Isolation sequence number | | |
| R.4.1.2 | Isolation test code | Isolation test code (if present) | |
| | ***Terminator record*** | L \| 1 \| N <CR> | Standard ASTM end |

### 3.2.2.3 Communication Example (message query result)

This is the answer to the patient query example from ***3.2.1.3***. After having sent the query message finished by an end-of-transmission (<EOT>) the MBT is waiting for an enquiry (<ENQ>) from the LIMS which starts the sending of the query result as shown here. Every sent ASTM frame from the LIMS is acknowledged by the MBT (<ACK>) until the final end-of-transmission (<EOT>) from the LIMS signals the channel to be closed.

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2P|1|81908|1957||CAMPA RODRIGUEZ^NIZAR^^^MR||19570227|M<CR><ETX>49<CR><LF>
<ACK>
<STX>3O|1|XY00012|||||201011301331||||AC 001||||||||||||||<CR><ETX>08<CR><LF>
<ACK>
<STX>4R|1|^^^ORG^1|vtekorg1<CR><ETX>FF<CR><LF>
<ACK>
<STX>5R|2|^^^ISOT^1|^T1<CR><ETX>08<CR><LF>
<ACK>
<STX>6L|1|N<CR><ETX>09<CR><LF>
<ACK>
<EOT>
```

## 3.3 Measuring Task Query

### 3.3.1 Query Message

#### 3.3.1.1 Message Structure

**Message header**

**Request Information (Target plate identifier)**

**Terminator**

#### 3.3.1.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | **H\|\^&<CR>** | Standard ASTM start |
| H.2 | Delimiters | **\|\^&** | Standard delimiters |
| | ***Request Information*** | **Q\|** Seq. No **\|** Instrument Id **^** Target plate Id **^** Project name **\|<CR>** | |
| Q.2.1.1 | Seq. No | Number of the Q-record | Here always 1 |
| Q.3.1.1 | Instrument Id | MALDI-TOF instrument identifier | Optional parameter |
| Q.3.1.2 | Target plate Id | MALDI-TOF target plate identifier | Required if no project name is provided |
| Q.3.1.3 | Project name | Measuring task project name | Required if no target plate Id is provided |
| | ***Terminator record*** | **L \| 1 \| N <CR>** | Standard ASTM end |

#### 3.3.1.3 Communication Example (message query)

This is a measuring task query using a target plate id (here: ***1010000007***). The first three digits of a target plate id specify the type (here: ***101***, meaning "MSP 96 target polished steel" target plate) while the following digits comprise the targets serial number (here: ***0000007***). As for all queries the MBT starts the communication with the enquiry (**<ENQ>**) followed by three ASTM frames acknowledged by the LIMS (**<ACK>**) and finished by the end-of-transmission (**<EOT>**) by the MBT. Again the answer of the query starts with an enquiry (**<ENQ>**) from the LIMS and is completely listed in paragraph ***3.3.2.3***.

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2Q|1|^1010000007^|<CR><ETX>DD<CR><LF>
<ACK>
<STX>3L|1|N<CR><ETX>06<CR><LF>
<ACK>
<EOT>
```

## 3.3.2 Query Result Message

### 3.3.2.1 Message Structrue

**Message header**

**Plate Information (from query)**

**[**

**Position Information (sample Id)**

**]**

**Terminator**

### 3.3.2.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | H\|\^&<CR> | Standard ASTM start |
| H.2 | Delimiters | \|\^& | Standard delimiters |
| | **Plate information** | O\| Seq. No \| Instrument Id ^ Target plate Id ^ Project name \|<CR> | |
| O.2.1.1 | Seq. No | Number of O-record | Starting at 1 |
| O.3.1.1 | Instrument Id | MALDI-TOF instrument identifier | Returned if provided |
| O.3.1.2 | Target plate Id | MALDI-TOF target plate identifier | Returned if provided |
| O.3.1.3 | Project name | Measuring task project name | Required parameter |
| | ***Position information*** | R\| Seq. No \| Accession number ^ Isolation number ^ Analyte name ^ Analyte description \|\|\|\| Target plate position ^ Target plate chip \|\|<CR> | |
| R.2.1.1 | Seq. No | Number of R-record | Starting at 1 |
| R.3.1.1 | Accession number | | Required parameter |
| R.3.1.2 | Isolation number | | Optional (default = 1) |
| R.3.1.3 | Analyte name | Name of analyte on target position | Optional parameter |
| R.3.1.4 | Analyte description | Description for analyte on target position | Optional parameter |
| R.4.1.1 | Analyte type (Sample type) | STD\|QC\|BC\|MYC\|FUNGI (standard \| quality control \| sepsityper \| mycobacteria \| filamentous fungi) | Optional parameter (default = **STD**, execept if R.3.1.1 is containing ***BTS*** – in this case **QC** is used) |
| R.5.1.1 | Preparation type | *(empty)*\|DT\|EDT\|EXT (unknown \| direct transfer \| extended direct transfer \| extraction) | Optional parameter (default is empty, this allows the LIMS to provide the preparation type for the MBT-Galaxy; used in MBT-Compass-HT) |
| R.7.1.1 | Target plate position | Position on MALDI target plate (e.g. A1) | Required parameter |
| R.7.1.2 | Target plate chip | Chip on MALDI target plate (0\|1\|2\|3) | Optional (default = 0) |
| | ***Terminator record*** | **L \| 1 \| N <CR>** | Standard ASTM end |

#### 3.3.2.3 Communication Example (message query result)

Here is the answer to the measuring task query example from ***3.3.1.3***. After having sent the query message finished by an end-of-transmission (**<EOT>**) the MBT is waiting for an enquiry (**<ENQ>**) from the LIMS which starts the sending of the query result as shown here. Every sent ASTM frame from the LIMS is acknowledged by the MBT (**<ACK>**) until the final end-of-transmission (**<EOT>**) from the LIMS signals the channel to be closed.

The returned measuring task contains the provided target plate id and a unique measuring task name (here ***140428-1437-DST-1010000007***) created by the LIMS (e.g. by prepending a time-stamp and a destination code to the target plate id). Following are twelve target position (***A1 – A12***) with the appropriate sample identifiers (please notice that a missing isolation number defaults to ***1*** while a missing sample type defaults to ***STD***).

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2O|1|^1010000007^140428-1437-DST-1010000007|<CR><ETX>38<CR><LF>
<ACK>
<STX>3R|1|XY00001^^name^BTS sample at A1|QC|||A1^||<CR><ETX>79<CR><LF>
<ACK>
<STX>4R|2|XY00002^1^^standard sample at A2|STD|||A2^||<CR><ETX>CD<CR><LF>
<ACK>
<STX>5R|3|XY00003^^name^sepsityper sample at A3|BC|||A3^||<CR><ETX>E3<CR><LF>
<ACK>
<STX>6R|4|XY00004^1^^standard sample at A4|STD|||A4^||<CR><ETX>D7<CR><LF>
<ACK>
<STX>7R|5|XY00005^^name^sepsityper sample at A5|BC|||A5^||<CR><ETX>ED<CR><LF>
<ACK>
<STX>0R|6|XY00006^1^^standard sample at A6|STD|||A6^||<CR><ETX>D9<CR><LF>
<ACK>
<STX>1R|7|XY00007^^name^sepsityper sample at A7|BC|||A7^||<CR><ETX>EF<CR><LF>
<ACK>
<STX>2R|8|XY00008^1^^standard sample at A8|STD|||A8^||<CR><ETX>E3<CR><LF>
<ACK>
<STX>3R|9|XY00009^^name^sepsityper sample at A9|BC|||A9^||<CR><ETX>F9<CR><LF>
<ACK>
<STX>4R|10|XY00010^^^standard sample (default) at A10||||A10^||<CR><ETX>93<CR><LF>
<ACK>
<STX>5R|11|XY00011^^name^sepsityper sample at A11|BC|||A11^||<CR><ETX>6F<CR><LF>
<ACK>
<STX>6R|12|XY00012^1^^mycobacteria sample at A12|MYC|||A12^||<CR><ETX>03<CR><LF>
<ACK>
<STX>7L|1|N<CR><ETX>0A<CR><LF>
<ACK>
<EOT>
```

## 3.4 Measuring Task List Query

### 3.4.1 Query Message

#### 3.4.1.1 Message Structure

**Message header**

**Request Information (Task name pattern)**

**Terminator**

#### 3.4.1.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | **H\|\^&<CR>** | Standard ASTM start |
| H.2 | Delimiters | **\|\^&** | Standard delimiters |
| | ***Request Information*** | **Q\|** Seq. No **\|\|\|\|\|** Task name pattern **^** Result count **\|\|\|\|\|<CR>** | |
| Q.2.1.1 | Seq. No | Number of the Q-record | Here always 1 |
| Q.7.1.1 | Task name pattern | Regular expression pattern like **.*?** for matching all possible names | Use ASTM escaping for coding delimiters |
| Q.7.1.2 | Result count | Number of results to be delivered (newest first) | All results if not present |
| | ***Terminator record*** | **L \| 1 \| N <CR>** | Standard ASTM end |

#### 3.4.1.3 Communication Example (message query)

This is a measuring task list query asking for all available measuring tasks (**.*?**). As usual the MBT starts the communication with the enquiry (**<ENQ>**) followed by three ASTM frames acknowledged by the LIMS (**<ACK>**) and finished by the end-of-transmission (**<EOT>**) by the MBT. Again the answer of the query starts with an enquiry (**<ENQ>**) from the LIMS and is completely listed in paragraph ***3.4.2.3***.

Please notice that according to the ASTM standard ***E 1394–97*** the delimiters (**|**,****,**^**,**&**) must be escaped if used in the regular expression pattern, like this: ***&F&***, ***&R&***, ***&S&***, ***&E&***, respectively.

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2Q|1|||||.*?^|||||<CR><ETX>0D<CR><LF>
<ACK>
<STX>3L|1|N<CR><ETX>06<CR><LF>
<ACK>
<EOT>
```

### 3.4.2 Query Result Message

#### 3.4.2.1 Message Structrue

**Message header**

**Pattern Information (from query)**

[

**Project Information (tasks)**

]

**Terminator**

### 3.4.2.2 Message Layout

| Component | | Syntax / Meaning | Comment |
|---|---|---|---|
| | ***Message Header*** | **H\|\^&<CR>** | Standard ASTM start |
| H.2 | Delimiters | **\|\^&** | Standard delimiters |
| | ***Pattern information*** | **O\|** Seq. No **\|** Task name pattern **\|<CR>** | |
| O.2.1.1 | Seq. No | Number of O-record | Starting at 1 |
| O.3.1.1 | Task pattern name | Matching pattern from query | |
| | ***Project information*** | **R\|** Seq. No. **\|** Project name **^** Project time-stamp **\|<CR>** | |
| R.2.1.1 | Seq. No | Number of the R-record | Here always 1 |
| R.3.1.1 | Project name | Measuring task project name | |
| R.3.1.2 | Project time-stamp | Creation time-stamp using this format **YYYY-MM-DD[T\| ]hh:mm:ss** | A **T** is preferred over a *blank* as separator between data and time |
| | ***Terminator record*** | **L\|1\|N <CR>** | Standard ASTM end |

### 3.4.2.3 Communication Example (message query result)

This is the answer to the measuring task list query example from ***3.4.1.3***. After having sent the query message finished by an end-of-transmission (**<EOT>**) the MBT is waiting for an enquiry (**<ENQ>**) from the LIMS which starts the sending of the query result as shown here. Every sent ASTM frame from the LIMS is acknowledged by the MBT (**<ACK>**) until the final end-of-transmission (**<EOT>**) from the LIMS signals the channel to be closed.

The returned measuring task list contains the provided pattern (***.*?***) and a list of three measuring task names together with their time-stamps.

```
<ENQ>
<ACK>
<STX>1H|\^&<CR><ETX>E5<CR><LF>
<ACK>
<STX>2O|1|.*?^|<CR><ETX>8E<CR><LF>
<ACK>
<STX>3R|1|Task001^2011-07-18T13:35:00|<CR><ETX>9E<CR><LF>
<ACK>
<STX>4R|2|Task002^2011-07-18T13:36:30|<CR><ETX>A5<CR><LF>
<ACK>
<STX>5R|3|Task003^2011-07-18T13:38:00|<CR><ETX>A7<CR><LF>
<ACK>
<STX>6L|1|N<CR><ETX>09<CR><LF>
<ACK>
<EOT>
```